IQRA UNIVERSITY

DEPARTMENT OF COMPUTER AND TECHNOLOGY

DLD LAB

LAB REPORT:02

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PRACTICAL WORK ON THE DIGITAL-ANALOG TRAINING SYSTEM

1. Objective

To understand the operation of a two-input AND gate using an AND-gate IC (assumed to be 7408), connect the IC on the trainer-board breadboard, apply logic inputs, observe the output on an LED/state monitor, and verify the truth table.

2. Apparatus

- IT-400 Digital-Analog Training System
- AND gate IC 7408
- Breadboard and connecting jumper wires
- +5 V DC supply and GND connection
- Logic switches / input terminals
- LED state monitor / output indicator

3. Theory

An AND gate is a basic digital logic gate. It gives logic 1 at its output only when both inputs are logic 1. If either input is 0, the output is 0. The Boolean expression is:

$$\mathbf{Y = A \cdot B}$$

The 7408 is a quad two-input AND-gate IC. It contains four independent AND gates. For the first gate of a standard 14-pin 7408, pins 1 and 2 are inputs, pin 3 is output, pin 14 is +5 V (VCC), and pin 7 is GND.

4. Procedure Performed on the Trainer Board

1. Identify the AND-gate IC and place it across the centre gap of the breadboard, ensuring that the IC pins are not shorted.
2. With the power OFF, connect pin 14 to +5 V and pin 7 to GND. Use the trainer board's regulated +5 V DC supply.
3. Connect input A and input B from two logic switches to pins 1 and 2 of the first AND gate.
4. Connect output pin 3 to an LED/state-monitor input so that the output can be observed.
5. Switch ON the trainer board. Apply the four possible input combinations (00, 01, 10, 11), one at a time.
6. Observe and record the output for each combination. The LED/state monitor represents logic 1 when it is ON and logic 0 when it is OFF (depending on the board's indicator convention).
7. Switch OFF the supply before changing any wiring.

5. Truth Table of AND Gate

Input A	Input B	Output Y = A · B	Expected LED
0	0	0	OFF
0	1	0	OFF
1	0	0	OFF
1	1	1	ON

6. Observation

The output was checked for all four input combinations. The AND gate produced logic 1 only when both inputs were logic 1. This agrees with the standard truth table of an AND gate. The uploaded photographs

show the IT-400 trainer board, breadboard, jumper-wire connections, and an illuminated state-monitor LED during the practical activity.

7. Conclusion

The AND gate IC was connected and tested on the IT-400 trainer board. By applying all four possible input combinations, its output behaviour was studied and compared with the truth table. The experiment demonstrated that an AND gate produces HIGH (1) only when both inputs are HIGH (1).

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DESIGN AND SIMULATION USING THREE LOGIC ICS

1. Objective

To design a combinational logic circuit using at least three logic ICs, simulate it in Multisim or Logisim, prepare its truth table, and verify the output.

2. ICs Used

IC Number	Gate Type	Purpose in the Circuit
7408	AND gate	Produces X = A · B
7404	NOT gate (inverter)	Produces $\overline{C}$ = NOT C
7432	OR gate	Combines X and $\overline{C}$ to produce final output F

3. Boolean Expression

$$F = (A \cdot B) + \overline{C}$$

The circuit has three inputs (A, B, C) and one output (F). First, A and B are connected to the AND gate. Input C is passed through the NOT gate. The two intermediate outputs are then connected to the OR gate.

4. Logic Circuit Diagram

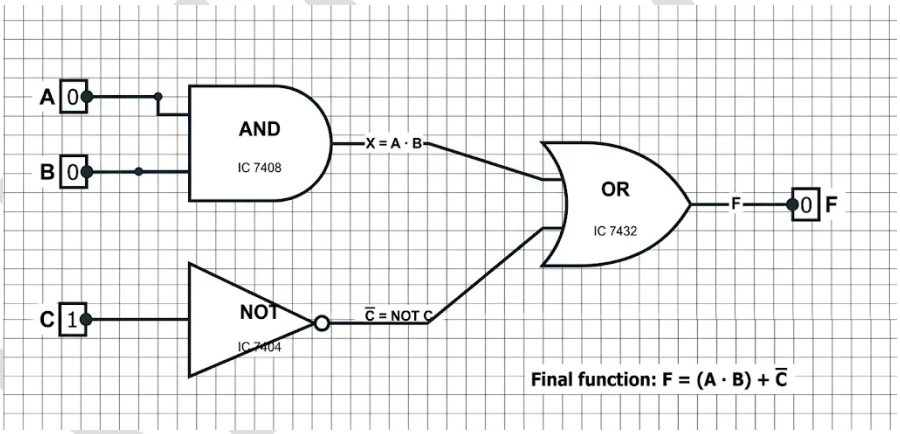


Figure 3: Logic diagram of F = (A · B) + $\overline{C}$ using ICs 7408, 7404 and 7432.

5. Pin Connections (Standard 14-pin DIP ICs)

IC	Pin Connections
7408 (AND)	Pin 1 = A; Pin 2 = B; Pin 3 = X; Pin 14 = +5 V; Pin 7 = GND
7404 (NOT)	Pin 1 = C; Pin 2 = $\overline{C}$; Pin 14 = +5 V; Pin 7 = GND
7432 (OR)	Pin 1 = X; Pin 2 = $\overline{C}$; Pin 3 = F; Pin 14 = +5 V; Pin 7 = GND

Important: These pin numbers are for the first gate/inverter of standard 14-pin 74xx DIP versions. Check the datasheet and the notch/pin-1 marking of the exact components before wiring. Connect the power pins of all

three ICs to the same regulated +5 V and common GND. Do not leave unused inputs floating; tie unused inputs to a defined logic level as recommended by the IC family.

6. Simulation Procedure (Multisim / Logisim)

8. Create a new digital circuit in Multisim or Logisim.
9. Place three ICs/gate packages: 7408 AND, 7404 NOT, and 7432 OR. If the software uses individual gates instead of IC packages, place the equivalent gates and label them with the IC numbers.
10. Add three logic input switches labelled A, B, and C, and one output LED/probe labelled F.
11. Wire A and B to the 7408 AND gate. Connect its output X to one input of the 7432 OR gate.
12. Wire C to the 7404 inverter. Connect its output $\overline{C}$ to the other input of the 7432 OR gate.
13. Connect the OR output to F. In Multisim, provide the required IC power connections if the selected component model exposes power pins.
14. Run the simulation and test all eight combinations of A, B, and C. Record the output in the truth table.

7. Truth Table

A	B	C	$\overline{C}$	A · B	Final F
0	0	0	1	0	1
0	0	1	0	0	0
0	1	0	1	0	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1

Note: F = (A · B) OR $\overline{C}$. Therefore, when C = 0, F is always 1. When C = 1, F equals A · B.

8. Expected Results

The output should be HIGH for input combinations 000, 010, 100, 110, and 111. It should be LOW for 001, 011, and 101. The simulated output should match every row of the truth table.

9. Conclusion

A combinational logic circuit was designed using three logic ICs: 7408 (AND), 7404 (NOT), and 7432 (OR). The circuit implements F = (A · B) + $\overline{C}$. Its truth table contains all eight possible combinations of three binary inputs. The output is HIGH whenever C is 0, or when A and B are both 1 while C is 1. The truth table provides the expected values for checking the Multisim/Logisim simulation.

10. Precautions

- Always switch OFF the power before inserting or removing ICs and changing connections.
- Check IC orientation and pin numbers before connecting wires.
- Use the correct regulated +5 V supply for standard 74xx logic ICs; never connect +12 V or -12 V to their supply pins.
- Ensure a common ground for the circuit and avoid short circuits between VCC and GND.
- Do not leave logic inputs floating; use logic switches or defined logic levels.
- Verify the IC part numbers printed on the components and use the matching datasheets.

Student note: This report is prepared from the supplied trainer-board photographs and the instructor's stated requirements. The photos do not clearly show the IC part number or a completed Multisim/Logisim simulation. Confirm that the physical IC was 7408, and build the three-IC circuit in your selected software

before submitting. Add your own simulation screenshot in this report if your instructor requires proof of simulation.

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